

T.C. Trakya University Electric-Electronic Engineering Department

EEM/EEE 229
Digital System Design

ASSIGNTMENT 1



Aydan Aydemir | 01.06.2026
1241604007

Assignment Details:

Students are required to design a 3-bit synchronous color counter circuit using D-type flip-flops. The circuit must cyclically and continuously repeat a specific state sequence determined according to the last five digits of the student ID number.

- The circuit shall use three D-type flip-flops: Q2, Q1, and Q0.

- State transitions shall occur only on the clock pulse.

- The circuit shall include eight colored LED outputs. For each state, only the corresponding color LED should be active.

- The circuit shall include one 7-segment display. According to the counter outputs Q2Q1Q0, the 7-segment display must indicate the current counter step from 0 to 7, while the color LEDs must operate separately according to the student's assigned color code sequence.

- The use of ready-made counters, predefined state machines, or ROMs is prohibited.

- The color output circuit must be designed by the student using the outputs Q2, Q1, and Q0.

DERIVATION OF THE COLOR CODE SEQUENCE:

The system architecture specifies a 3-bit synchronous color counter that cycles through eight distinct states (0 to 7) continuously on successive clock pulses. The sequence of colors displayed by the eight output LEDs is uniquely determined by the student ID number, combining three fixed initial steps with five variable steps extracted from the final digits of the ID.

- Student ID Number: 1241604007
- Extracted Last Five Digits: 0, 4, 0, 0, 7

Digit Interpretation Rules:

1. Digits from 0 to 7 are utilized directly without any modification.
2. The digit 8 is interpreted as 0 (not applicable for my ID).
3. The digit 9 is interpreted as 1 (not applicable for my ID).

Sequence Assembly:

- Steps 0 to 2 (Fixed for all students): 4,6,1
- Steps 3 to 7 (Derived from ID 0, 4, 0, 0, 7): 0,4,0,0,7

Combining these rules, the complete, cyclical 8-step state sequence mapped to the 3-bit counter output (Q_2 Q_1 Q_0) is detailed below:

State Number	**Binary State ($Q_2Q_1Q_0$)**	**Color Code**	**Corresponding Color LED Output**
0	000	4	Blue
1	001	6	Pink
2	010	1	Orange
3	011	0	Red
4	100	4	Blue
5	101	0	Red
6	110	0	Red
7	111	7	White

The generated color code sequence to be repeated cyclically and indefinitely by the hardware is:
4→6→1→0→4→0→0→7

STATE TRANSITION TABLE AND D FLIP-FLOP INPUT TABLE

The system utilizes three D-type flip-flops designated as Q_2 (Most Significant Bit - MSB), Q_1, and Q_0 (Least Significant Bit - LSB). For a D-type flip-flop, the excitation equation dictating the next state is given by:

$Q^+ = D$

Therefore, the input required at each flip-flop (D_2, D_1, D_0) directly corresponds to the desired next-state value (Q^+_2, Q^+_1, Q^+_0).

Since the 7-segment display must indicate the current step sequentially from 0 to 7, the counter is modeled as a standard 3-bit binary up-counter.

Current State (Q2Q1 Q0)	Decimal	Next State (Q2+Q1+ Q0+)	Flip-Flop Inputs (D2D1 D0)
0 0 0	0	0 0 1	0 0 1
0 0 1	1	0 1 0	0 1 0
0 1 0	2	0 1 1	0 1 1
0 1 1	3	1 0 0	1 0 0
1 0 0	4	1 0 1	1 0 1
1 0 1	5	1 1 0	1 1 0
1 1 0	6	1 1 1	1 1 1
1 1 1	7	0 0 0	0 0 0

STATE TRANSITION TABLE AND D FLIP-FLOP INPUT TABLE

The circuit utilizes three D-type flip-flops designated as Q2 (MSB), Q1, and Q0 (LSB). For D-type flip-flops, the next-state value equals the input ($Q^+ = D$). The counter counts sequentially from 0 to 7 as a standard binary up-counter.

Current State (Q2 Q1 Q0)	Decimal	Next State (Q2+ Q1+ Q0+)	Flip-Flop Inputs (D2 D1 D0)
0 0 0	0	0 0 1	0 0 1
0 0 1	1	0 1 0	0 1 0
0 1 0	2	0 1 1	0 1 1
0 1 1	3	1 0 0	1 0 0
1 0 0	4	1 0 1	1 0 1
1 0 1	5	1 1 0	1 1 0
1 1 0	6	1 1 1	1 1 1
1 1 1	7	0 0 0	0 0 0

TRUTH TABLES, KARNAUGH MAPS, AND SIMPLIFIED BOOLEAN EXPRESSIONS

This section details the combinational logic synthesis for both the internal counter execution and the custom 7-segment display driver.

Counter Input Logic Synthesis

Q1Q0 \ Q2	Q2 = 0	Q2 = 1
00	1	1
01	0	0
11	0	0
10	1	1

Q1Q0 \ Q2	Q2 = 0	Q2 = 1
00	0	0
01	1	1
11	0	0
10	1	1

Q1Q0 \ Q2	Q2 = 0	Q2 = 1
00	0	1
01	0	1
11	1	0
10	0	1

The minimized Boolean expressions for the D flip-flop inputs obtained from Karnaugh maps are as follows:

- Expression for D_0: $D_0 = Q_0'$
- Expression for D_1: $D_1 = Q_1'Q_0 + Q_1Q_0' = Q_1 \oplus Q_0$
- Expression for D_2: $D_2 = Q_2Q_1' + Q_2Q_0' + Q_2Q_1Q_0 = Q_2 \oplus (Q_1\ Q_0)$

7-Segment Display Driver Design

To display steps 0 to 7 without using ready-made decoder ICs, a manual active-high (common cathode) 7-segment driver was synthesized.

Q2	Q1	Q0	Decimal	a	b	c	d	e	f	g
0	0	0	0	1	1	1	1	1	1	0
0	0	1	1	0	1	1	0	0	0	0
0	1	0	2	1	1	0	1	1	0	1
0	1	1	3	1	1	1	1	0	0	1
1	0	0	4	0	1	1	0	0	1	1
1	0	1	5	1	0	1	1	0	1	1
1	1	0	6	1	0	1	1	1	1	1
1	1	1	7	1	1	1	0	0	0	0

a = Q1 + (Q2 * Q0) + (Q2' * Q0')

b = Q2' + (Q1' * Q0') + (Q1 * Q0)

c = Q2 + Q1' + Q0

d = (Q2' * Q0') + (Q2' * Q1) + (Q1 * Q0') + (Q2 * Q1' * Q0)

e = (Q2' * Q0') + (Q1 * Q0')

f = (Q2 * Q1') + (Q1' * Q0') + (Q2 * Q0')

g = (Q2 * Q1') + (Q2' * Q1) + (Q2 * Q0)

Truth Table and Boolean Expressions for the Color LED Outputs

The combinational logic truth table mapping the counter states to the student-specific color outputs is detailed below:

Q2	Q1	Q0	Active Color	Red	Orange	Yellow	Green	Blue	Purple	Pink	White
0	0	0	Blue	0	0	0	0	1	0	0	0
0	0	1	Pink	0	0	0	0	0	0	1	0
0	1	0	Orange	0	1	0	0	0	0	0	0
0	1	1	Red	1	0	0	0	0	0	0	0
1	0	0	Blue	0	0	0	0	1	0	0	0
1	0	1	Red	1	0	0	0	0	0	0	0
1	1	0	Red	1	0	0	0	0	0	0	0
1	1	1	White	0	0	0	0	0	0	0	1

Simplified Boolean Expressions for Color LEDs:

Blue = Q1' * Q0'

Pink = Q2' * Q1' * Q0

Orange = Q2' * Q1 * Q0'

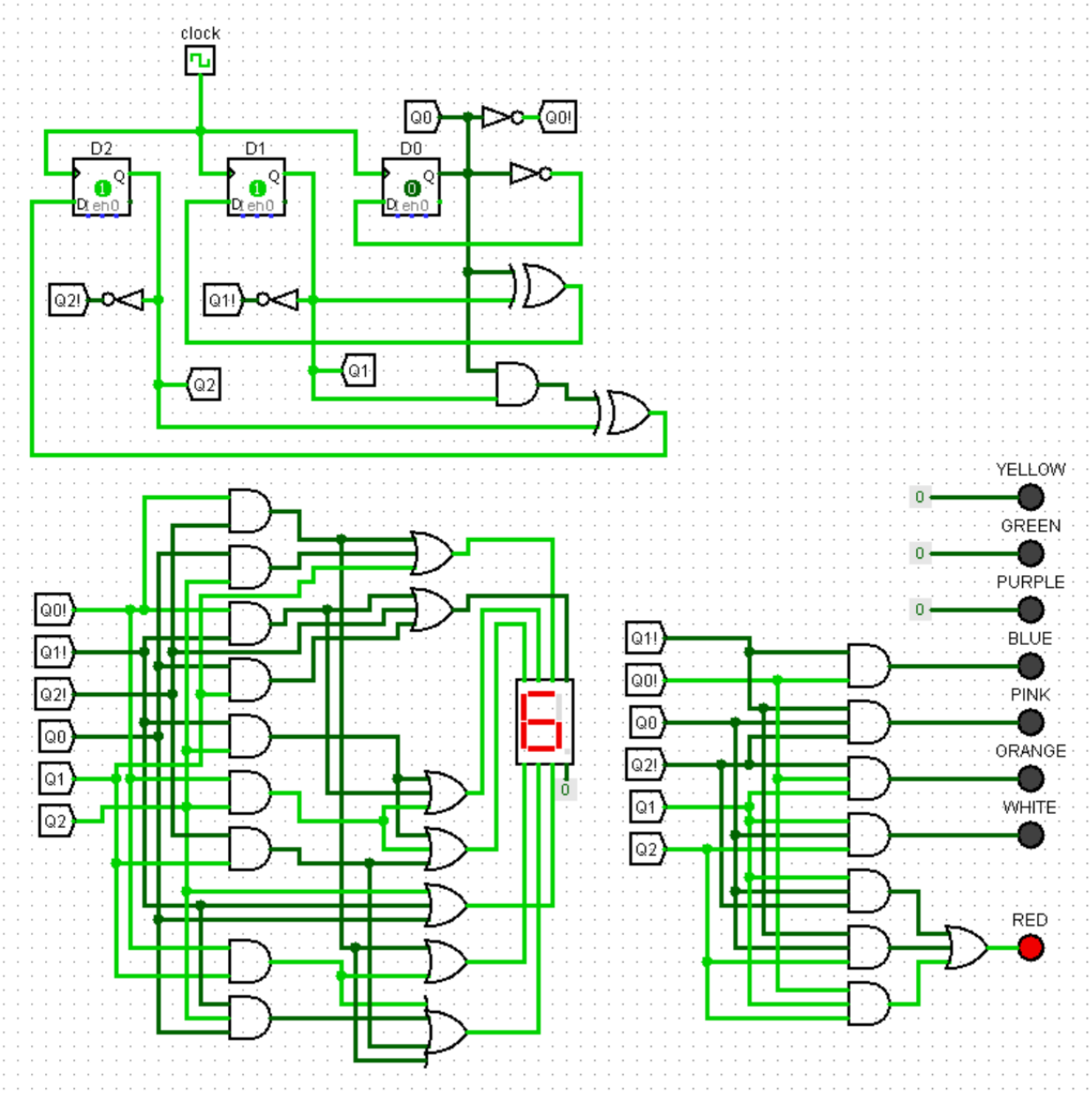
Red = (Q2' * Q1 * Q0) + (Q2 * Q1' * Q0) + (Q2 * Q1 * Q0')

White = Q2 * Q1 * Q0

Yellow = 0, Green = 0, Purple = 0 (Grounded channels)

SCREENSHOT OF THE CIRCUIT DESIGN IMPLEMENTED IN LOGISIM

The schematic layout designed within Logisim is segmented into three logical zones. The upper section contains the synchronous 3-bit counter engine driven by a single clock element. Signal serialization is optimized using global net labels (Tunnels) to decouple the memory elements from the decoder stages. The lower-left area accommodates the custom gate arrays processing segments a through g, routing directly into the standard 7-Segment Display component. The lower-right network holds the independent LED decoding matrix controlling the physical status indicators.



THE VERILOG CODE AND A SCREENSHOT OF THE RTL SCHEMATIC

The system hardware configuration was described structurally and behaviorally using Verilog HDL. The module is declared as `eee229` to match the mandatory top-level design entity parameters mapped within Intel Quartus.

```
// ================================================================================
// Course: Digital System Design (EEE/EEM 229)
// Description: 3-Bit Synchronous Color Counter and 7-Segment Driver
// Top-Level Entity Match: eee229
// ================================================================================

module eee229 (
    input clk,          // Shared Clock Signal
    output Q2, Q1, Q0,  // 3-bit Counter Outputs
    output Red, Orange, Yellow, Green, Blue, Purple, Pink, White, // LED Array
    output a, b, c, d, e, f, g // 7-Segment Display Line Array
);

    // Internal State Registers for D Flip-Flops
    reg q2_reg = 0;
    reg q1_reg = 0;
    reg q0_reg = 0;

    // Excitation Wiring Nets
    wire d2, d1, d0;

    // --------------------------------------------------------------------------
    // 1. Counter Next-State Logic Realization
    // --------------------------------------------------------------------------
    assign d0 = ~q0_reg;
    assign d1 = q1_reg ^ q0_reg;
    assign d2 = q2_reg ^ (q1_reg & q0_reg);

    // Synchronous State Transition Edge-Triggered Block
    always @(posedge clk) begin
        q0_reg <= d0;
        q1_reg <= d1;
        q2_reg <= d2;
    end

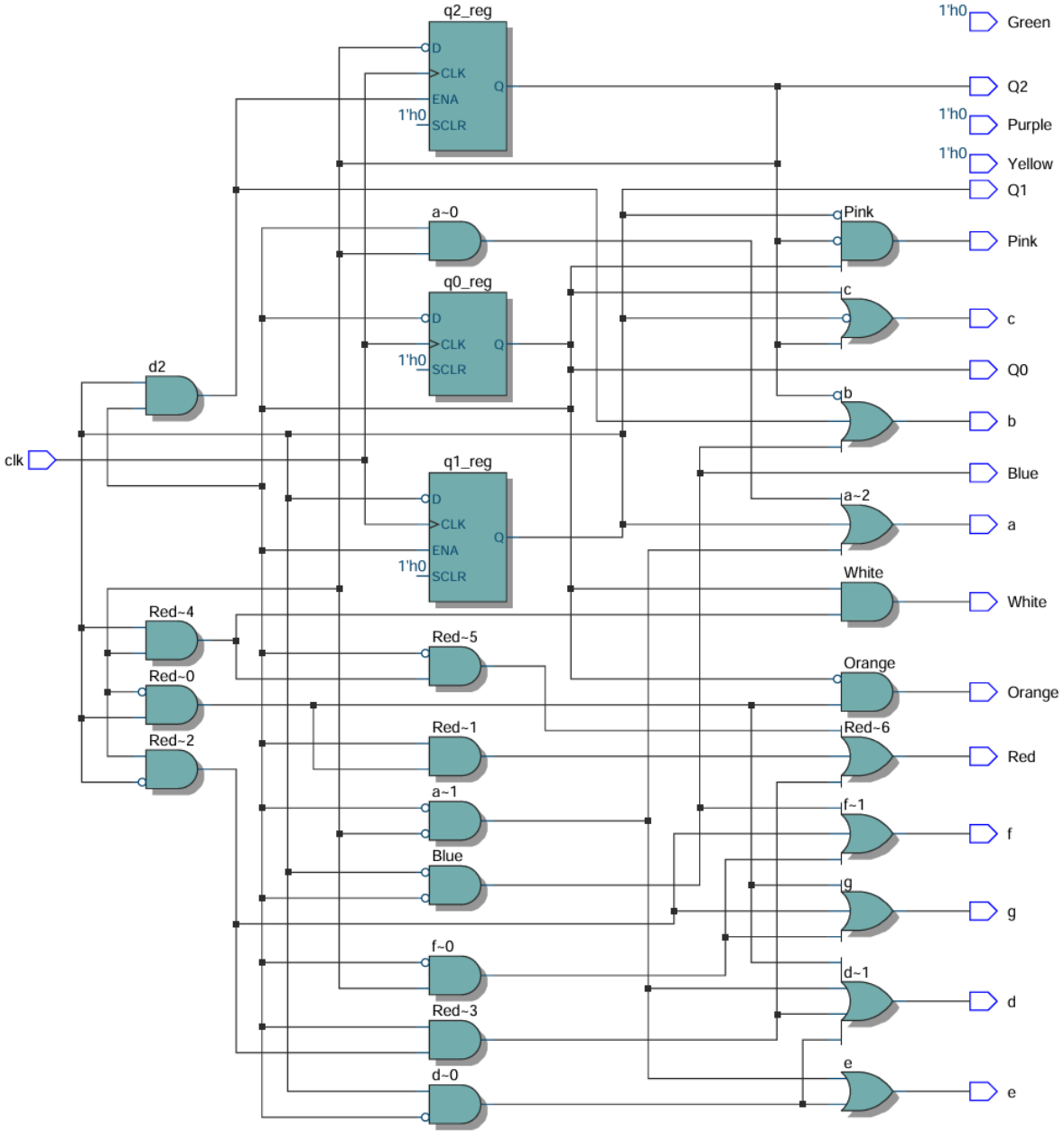
    // Direct Port Boundary Mapping
    assign Q2 = q2_reg;
    assign Q1 = q1_reg;
    assign Q0 = q0_reg;

    // --------------------------------------------------------------------------
    // 2. Color Mapping Combinational Logic
    // --------------------------------------------------------------------------
    assign Blue   = (~Q1 & ~Q0);
    assign Pink   = (~Q2 & ~Q1 & Q0);
    assign Orange = (~Q2 & Q1 & ~Q0);
    assign Red    = (~Q2 & Q1 & Q0) | (Q2 & ~Q1 & Q0) | (Q2 & Q1 & ~Q0);
    assign White  = (Q2 & Q1 & Q0);

    // Unused Color Channels Grounded
    assign Yellow = 1'b0;
    assign Green  = 1'b0;
    assign Purple = 1'b0;

    // --------------------------------------------------------------------------
    // 3. 7-Segment Custom Combinational Driver
    // --------------------------------------------------------------------------
    assign a = Q1 | (Q2 & Q0) | (~Q2 & ~Q0);
    assign b = ~Q2 | (~Q1 & ~Q0) | (Q1 & Q0);
    assign c = Q2 | ~Q1 | Q0;
    assign d = (~Q2 & ~Q0) | (~Q2 & Q1) | (Q1 & ~Q0) | (Q2 & ~Q1 & Q0);
    assign e = (~Q2 & ~Q0) | (Q1 & ~Q0);
    assign f = (Q2 & ~Q1) | (~Q1 & ~Q0) | (Q2 & ~Q0);
    assign g = (Q2 & ~Q1) | (~Q2 & Q1) | (Q2 & ~0);

endmodule
```


q2_reg
D
CLK
ENA
SCLR
Q
1'h0
q0_reg
q1_reg
d2
clk
Red~4
Red~0
Red~2
a~0
Red~5
Red~1
a~1
Blue
f~0
Red~3
d~0
Pink
c
b
a~2
White
Orange
Red~6
f~1
g
d~1
e
1'h0
Green
Q2
Purple
Yellow
Q1
Pink
c
Q0
b
Blue
a
White
Orange
Red
f
g
d
e